K.R. MANGALAM UNIVERSITY
B.TECH COMPUTER SCIENCE (2ND YEAR)
END-TERM EXAMINATION QUESTION BANK

Unit 1: Differential Equations (1st & 2nd Order)

ORDER & DEGREE *[2 Marks]*

Q: Determine the order and degree of: $(d^2y/dx^2)^3 + 4(dy/dx)^4 + y = \sin(x)$.

Step 1: Identify the highest derivative. The highest order derivative is d^2y/dx^2.

Step 2: The order is the highest derivative present (2).

Step 3: The degree is the highest power of the highest order derivative (3).

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Answer: Order = 2, Degree = 3
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Unit 1

TYPE: LINEAR DE *[10 Marks]*

Q: Solve the linear differential equation: $dy/dx + (2/x)y = x^2$.

Step 1: Identify standard form $dy/dx + Py = Q$. Here, $P = 2/x$, $Q = x^2$.

Step 2: Calculate Integrating Factor (I.F) = $e^{\int P\,dx} = e^{\int (2/x)dx} = e^{2 \ln x} = x^2$.

Step 3: Solution is $y * (\text{I.F}) = \int Q * (\text{I.F})\, dx + C$.

Step 4: $y(x^2) = \int (x^2)(x^2)\, dx = \int x^4\, dx = x^5/5 + C$.

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Answer: y = x³/5 + C/x²
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TYPE: EXACT DE *[10 Marks]*

Q: Solve the exact equation: $(2xy + 3)dx + (x^2 - 1)dy = 0$.

Step 1: Let $M = 2xy + 3$, $N = x^2 - 1$. Check exactness: $\partial M/\partial y = 2x$, $\partial N/\partial x = 2x$. (Exact).

Step 2: $\int M\,dx$ (y constant) = $\int(2xy + 3)dx = x^2y + 3x$.

Step 3: $\int$ (Terms of N without x) dy = $\int(-1)dy = -y$.

Step 4: Add results and equate to C.

Answer: $x^2y + 3x - y = C$

TYPE: HOMOGENEOUS DE

[10 Marks]

Q: Solve the homogeneous equation: $(x^2 + y^2)dx - 2xy\,dy = 0$.

Step 1: Rewrite as $dy/dx = (x^2 + y^2) / 2xy$. Substitute $y = vx$, $dy/dx = v + x(dv/dx)$.

Step 2: $v + x(dv/dx) = (x^2 + v^2x^2) / 2x^2v = (1 + v^2) / 2v$.

Step 3: $x(dv/dx) = (1 + v^2)/2v - v = (1 - v^2) / 2v$.

Step 4: Separate variables: $\int(2v/(1-v^2))dv = \int(1/x)dx$. Let $u = 1-v^2$, $du = -2v\,dv$. $-\ln(1-v^2) = \ln(x) + C$.

Step 5: $\ln(1/(1-v^2)) = \ln(Cx)$ -> $1 - (y/x)^2 = C/x$.

Answer: $x^2 - y^2 = Cx$

TYPE: SEPARABLE DE

[10 Marks]

Q: Solve the separable equation: dy/dx = (x e^x) / y.

Step 1: Separate variables: y dy = x e^x dx.

Step 2: Integrate both sides: ∫y dy = ∫x e^x dx.

Step 3: LHS = $y^2/2$. RHS uses integration by parts (u=x, dv=e^x dx): x e^x - ∫e^x dx = x e^x - e^x.

Step 4: $y^2/2$ = e^x(x - 1) + C.

Answer: y^2 = 2e^x(x - 1) + C

TYPE: BERNOULLI DE

[10 Marks]

Q: Solve Bernoulli's equation: $dy/dx + y = y^2$.

Step 1: Divide by y^2: $y^{-2}(dy/dx) + y^{-1} = 1$.

Step 2: Let $v = y^{-1}$. Then $dv/dx = -y^{-2}(dy/dx)$. Substitute: $-dv/dx + v = 1$ => $dv/dx - v = -1$.

Step 3: Linear in v. I.F = e^(∫-1 dx) = e^{-x}.

Step 4: $v(e^{-x}) = \int(-1)(e^{-x})dx = e^{-x} + C$. Therefore, $v = 1 + C e^{x}$.

Step 5: Substitute back $y = 1/v$.

Answer: $y = 1 / (1 + C e^{x})$

INITIAL VALUE PROBLEM

[10 Marks]

Q: Solve the IVP: $dy/dx = -2xy$, with $y(0) = 5$.

Step 1: Separate variables: $dy/y = -2x\, dx$.

Step 2: Integrate: $\ln|y| = -x^2 + C \Rightarrow y = C e^{(-x^2)}$.

Step 3: Apply initial condition $y(0) = 5$. $5 = C e^{(0)} \Rightarrow C = 5$.

Step 4: Substitute C back into general solution.

Answer: $y = 5 e^{(-x^2)}$

BOUNDARY VALUE PROBLEM

[10 Marks]

Q: Solve the BVP: $y'' + 4y = 0$, $y(0) = 0$, $y(\pi/4) = 2$.

Step 1: Characteristic equation $r^2 + 4 = 0 \Rightarrow r = \pm 2i$.

Step 2: General solution: $y(x) = C_1 \cos(2x) + C_2 \sin(2x)$.

Step 3: Apply $y(0) = 0$: $0 = C_1(1) + C_2(0) \Rightarrow C_1 = 0$. So $y(x) = C_2 \sin(2x)$.

Step 4: Apply $y(\pi/4) = 2$: $2 = C_2 \sin(\pi/2) \Rightarrow 2 = C_2(1) \Rightarrow C_2 = 2$.

Answer: $y(x) = 2 \sin(2x)$

2ND ORDER HOMOGENEOUS

[10 Marks]

Q: Solve: $d^2y/dx^2 - 5(dy/dx) + 6y = 0$.

Step 1: Write the auxiliary equation: $m^2 - 5m + 6 = 0$.

Step 2: Factorize: $(m - 2)(m - 3) = 0$. Roots are $m_1 = 2$, $m_2 = 3$.

Step 3: Since roots are real and distinct, the solution takes the form $y = C_1 e^{(m_1 x)} + C_2 e^{(m_2 x)}$.

Answer: $y = C_1 e^{(2x)} + C_2 e^{(3x)}$

2ND ORDER NON-HOMOGENEOUS

[10 Marks]

Q: Find the general solution of $y'' - 4y = e^{3x}$.

Step 1: Find Complementary Function (C.F). $m^2 - 4 = 0 \Rightarrow m = \pm 2$. C.F = $C_1 e^{2x} + C_2 e^{-2x}$.

Step 2: Find Particular Integral (P.I). P.I = $1/(D^2 - 4) [e^{3x}]$.

Step 3: Replace D with a=3. P.I = $1/(3^2 - 4) * e^{3x} = 1/(9 - 4) * e^{3x} = (1/5)e^{3x}$.

Step 4: General Solution y = C.F + P.I.

Answer: $y = C_1 e^{2x} + C_2 e^{-2x} + (1/5)e^{3x}$

Unit 2: Numerical Methods

ERROR ANALYSIS

[2 Marks]

Q: Define Absolute, Relative, and Percentage Error.

Step 1: Absolute Error (Ea) = |True Value - Approximate Value|.

Step 2: Relative Error (Er) = Absolute Error / |True Value|.

Step 3: Percentage Error (Ep) = Relative Error × 100%.

Answer: Formulas defined in steps.

Unit 2

BISECTION METHOD

[10 Marks]

Q: Find the root of $x^3 - x - 1 = 0$ lying between 1 and 2, using 2 iterations.

Step 1: f(1) = 1 - 1 - 1 = -1 (Negative). f(2) = 8 - 2 - 1 = 5 (Positive). Root lies in [1, 2].

Step 2: Iteration 1. Midpoint $x_1 = (1+2)/2 = 1.5$. $f(1.5) = 1.5^3 - 1.5 - 1 = 0.875$ (Positive). Root in [1, 1.5].

Step 3: Iteration 2. Midpoint $x_2 = (1+1.5)/2 = 1.25$. $f(1.25) = 1.25^3 - 1.25 - 1 = -0.296$ (Negative).

Answer: Root is approximately 1.25

SECANT METHOD

[10 Marks]

Q: Find the root of $x^2 - 4 = 0$ using Secant Method with initial guesses $x_0 = 1$, $x_1 = 3$ (1 iteration).

> Step 1: $f(x_0) = f(1) = 1 - 4 = -3$. $f(x_1) = f(3) = 9 - 4 = 5$.
>
> Step 2: Formula: $x_2 = x_1 - f(x_1)[(x_1 - x_0) / (f(x_1) - f(x_0))]$.
>
> Step 3: $x_2 = 3 - 5 * [(3 - 1) / (5 - (-3))] = 3 - 5 * [2 / 8] = 3 - 5(0.25) = 3 - 1.25 = 1.75$.
>
> **Answer: $x_2 = 1.75$**

NEWTON RAPHSON METHOD

[10 Marks]

Q: Solve $x^3 - 2x - 5 = 0$ near $x_0 = 2$ using one iteration.

> Step 1: $f(x) = x^3 - 2x - 5$. Derivative $f'(x) = 3x^2 - 2$.
>
> Step 2: Evaluate at $x_0 = 2$. $f(2) = 8 - 4 - 5 = -1$. $f'(2) = 12 - 2 = 10$.
>
> Step 3: Apply formula: $x_1 = x_0 - f(x_0)/f'(x_0) = 2 - (-1)/10 = 2 + 0.1 = 2.1$.
>
> **Answer: $x_1 = 2.1$**

REGULA FALSI METHOD

[10 Marks]

Q: Find a root of $x^2 - 3 = 0$ in [1, 2] using 1 iteration.

> Step 1: $a=1$, $f(a) = -2$. $b=2$, $f(b) = 1$.
>
> Step 2: Formula $x_1 = (a*f(b) - b*f(a)) / (f(b) - f(a))$.
>
> Step 3: $x_1 = (1*1 - 2*(-2)) / (1 - (-2)) = (1 + 4) / 3 = 5/3 \approx 1.666$.
>
> **Answer: $x_1 = 1.666$**

LU DECOMPOSITION

[10 Marks]

Q: Decompose matrix A = [[2, 3], [4, 7]] into L and U matrices.

> Step 1: Let $A = LU$. $L = [[1, 0], [L_{21}, 1]]$, $U = [[U_{11}, U_{12}], [0, U_{22}]]$.
>
> Step 2: Multiply $LU = [[U_{11}, U_{12}], [L_{21}U_{11}, L_{21}U_{12} + U_{22}]]$.

Step 3: Equate to A. $U_{11} = 2$, $U_{12} = 3$.

Step 4: $L_{21}U_{11} = 4 \Rightarrow L_{21}(2) = 4 \Rightarrow L_{21} = 2$.

Step 5: $L_{21}U_{12} + U_{22} = 7 \Rightarrow 2(3) + U_{22} = 7 \Rightarrow 6 + U_{22} = 7 \Rightarrow U_{22} = 1$.

Answer: L = [[1, 0], [2, 1]], U = [[2, 3], [0, 1]]

SVD

[10 Marks]

Q: Outline the analytical steps to find the Singular Value Decomposition (SVD) of a matrix A.

Step 1: Compute the matrix A^T A and A A^T.

Step 2: Find the eigenvalues and eigenvectors of A^T A. The eigenvectors form the columns of V.

Step 3: The square roots of the non-zero eigenvalues are the singular values, placed diagonally in matrix Σ.

Step 4: Find the eigenvectors of A A^T to form the orthogonal matrix U. Assemble A = U Σ V^T.

Answer: A = U Σ V^T (Steps outlined)

GAUSS-SEIDEL METHOD

[10 Marks]

Q: Solve: 4x + y = 5, x + 3y = 4. Initial guess (0,0). 1 iteration.

Step 1: Rearrange for strict diagonal dominance. $x = (5 - y)/4$. $y = (4 - x)/3$.

Step 2: Use most recent values immediately. $x_1 = (5 - 0)/4 = 1.25$.

Step 3: $y_1 = (4 - x_1)/3 = (4 - 1.25)/3 = 2.75 / 3 \approx 0.916$.

Answer: $x_1 = 1.25$, $y_1 = 0.916$

JACOBI METHOD

[10 Marks]

Q: Solve: 4x + y = 5, x + 3y = 4. Initial guess (0,0). 1 iteration.

Step 1: Rearrange: $x = (5 - y)/4$, $y = (4 - x)/3$.

Step 2: Use old values only for the iteration. $x_1 = (5 - y_0)/4 = 5/4 = 1.25$.

Step 3: $y_1 = (4 - x_0)/3 = 4/3 \approx 1.333$.

Answer: $x_1 = 1.25$, $y_1 = 1.333$

CHOLESKY DECOMPOSITION

[10 Marks]

Q: Find Cholesky decomposition of A = [[4, 2], [2, 10]].

Step 1: Let A = LL^T. L = [[L_{11}, 0], [L_{21}, L_{22}]]. LL^T = [[L_{11}^2, $L_{11}L_{21}$], [$L_{11}L_{21}$, $L_{21}^2 + L_{22}^2$]].

Step 2: $L_{11}^2 = 4 \Rightarrow L_{11} = 2$.

Step 3: $L_{11}L_{21} = 2 \Rightarrow 2*L_{21} = 2 \Rightarrow L_{21} = 1$.

Step 4: $L_{21}^2 + L_{22}^2 = 10 \Rightarrow 1^2 + L_{22}^2 = 10 \Rightarrow L_{22}^2 = 9 \Rightarrow L_{22} = 3$.

Answer: L = [[2, 0], [1, 3]]

FINITE DIFFERENCE

[10 Marks]

Q: Construct a forward difference table for $y = x^2$ for x = 1, 2, 3.

Step 1: Calculate y values. y(1)=1, y(2)=4, y(3)=9.

Step 2: First differences (Δy): $\Delta y_1 = 4 - 1 = 3$. $\Delta y_2 = 9 - 4 = 5$.

Step 3: Second differences ($\Delta^2 y$): $\Delta^2 y_1 = 5 - 3 = 2$.

Answer: Δy = [3, 5], $\Delta^2 y$ = [2]

Unit 3: Random Variables

DISCRETE VS CONTINUOUS

[2 Marks]

Q: State the key difference between Discrete and Continuous Random Variables.

Step 1: A discrete random variable takes on a countable number of distinct values (e.g., number of students).

Step 2: A continuous random variable takes on an infinite number of possible values within a given interval (e.g., exact height or weight).

Answer: Countable vs. Infinite Interval Values

Unit 3

MEAN, VARIANCE, STANDARD DEVIATION

[10 Marks]

Q: A random variable X has P(0)=0.2, P(1)=0.5, P(2)=0.3. Find Mean, Variance, and SD.

> Step 1: Mean (E[X]) = $\Sigma x_i P(x_i)$ = 0(0.2) + 1(0.5) + 2(0.3) = 0 + 0.5 + 0.6 = 1.1.
>
> Step 2: $E[X^2] = \Sigma x_i^2 P(x_i)$ = 0(0.2) + 1(0.5) + 4(0.3) = 0.5 + 1.2 = 1.7.
>
> Step 3: Variance (Var) = $E[X^2] - (E[X])^2 = 1.7 - (1.1)^2 = 1.7 - 1.21 = 0.49$.
>
> Step 4: Standard Deviation (SD) = $\sqrt{Var} = \sqrt{0.49} = 0.7$.
>
> **Answer: Mean = 1.1, Var = 0.49, SD = 0.7**

CENTRAL LIMIT THEOREM

[10 Marks]

Q: Define the Central Limit Theorem.

> Step 1: For a population with mean μ and variance σ^2...
>
> Step 2: ...the distribution of sample means of size n approaches a normal distribution as n becomes large ($n \geq 30$).
>
> Step 3: The new normal distribution will have mean μ and variance σ^2/n.
>
> **Answer: $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n}) \sim N(0,1)$**

BERNOULLI DISTRIBUTION

[2 Marks]

Q: Give the probability mass function (PMF) and mean for a Bernoulli distribution.

> Step 1: PMF: $P(X=x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$.
>
> Step 2: Mean (E[X]) = p. Variance = p(1-p).
>
> **Answer: PMF = $p^x(1-p)^{1-x}$, Mean = p**

BINOMIAL DISTRIBUTION

[10 Marks]

Q: A fair coin is tossed 4 times. Find the probability of getting exactly 2 heads.

> Step 1: Parameters: n=4, p=0.5 (probability of head), r=2.
>
> Step 2: Formula: $P(X=r) = {}^nC_r\, p_r\, q^{n-}{}_r$.
>
> Step 3: $P(X=2) = {}^4C_2\, (0.5)^2\, (0.5)^2 = 6 * 0.25 * 0.25 = 6/16 = 3/8$.
>
> **Answer: P(X=2) = 0.375**

POISSON DISTRIBUTION *[10 Marks]*

Q: A call center receives an average of 3 calls per minute. Find the probability of exactly 1 call in a given minute.

> Step 1: Identify Poisson parameter $\lambda = 3$.
>
> Step 2: Formula: $P(X=x) = (e^{-}\lambda * \lambda^x) / x!$
>
> Step 3: $P(X=1) = (e^{-3} * 3^1) / 1! = 3 / e^3 \approx 3 / 20.08 \approx 0.149$.
>
> **Answer: P(X=1) ≈ 0.149**

EXPONENTIAL DISTRIBUTION *[10 Marks]*

Q: The life of a bulb follows an exponential distribution with mean 100 hours. Find the probability it fails before 50 hours.

> Step 1: Mean = $1/\lambda = 100$, so $\lambda = 0.01$.
>
> Step 2: CDF Formula for $P(X < x) = 1 - e^{(-\lambda x)}$.
>
> Step 3: $P(X < 50) = 1 - e^{(-0.01 * 50)} = 1 - e^{(-0.5)}$.
>
> Step 4: 1 - 0.6065 = 0.3935.
>
> **Answer: P(X < 50) ≈ 39.35%**

Unit 4: Optimization

CONVEX SET & FUNCTION *[2 Marks]*

Q: Define a Convex Set and a Convex Function.

> Step 1: Convex Set: A set where a line segment connecting any two points in the set lies entirely within the set.
>
> Step 2: Convex Function: A function where the line segment between any two points on its graph lies above or on the graph.
>
> **Answer: Definitions provided in steps.**

Unit 4

OPTIMISATION *[10 Marks]*

Q: Find the maximum or minimum value of $f(x) = x^2 - 6x + 8$.

> Step 1: Find first derivative: $f'(x) = 2x - 6$.
>
> Step 2: Set $f'(x) = 0$ for critical points. $2x - 6 = 0 \Rightarrow x = 3$.
>
> Step 3: Find second derivative to test: $f''(x) = 2$.
>
> Step 4: Since $f''(3) = 2 > 0$, the function has a local minimum at $x = 3$.
>
> Step 5: Minimum value $= f(3) = 3^2 - 6(3) + 8 = 9 - 18 + 8 = -1$.
>
> **Answer: Minimum value is -1 at x = 3**

LAGRANGE MULTIPLIER *[10 Marks]*

Q: Maximize $f(x,y) = xy$ subject to $x + y = 10$.

> Step 1: Define Lagrangian $L(x,y,\lambda) = xy - \lambda(x + y - 10)$.
>
> Step 2: Partial derivatives: $\partial L/\partial x = y - \lambda = 0$, $\partial L/\partial y = x - \lambda = 0$, $\partial L/\partial \lambda = -(x + y - 10) = 0$.
>
> Step 3: From first two, $y = \lambda$ and $x = \lambda$, so $x = y$.
>
> Step 4: Substitute into constraint: $x + x = 10 \Rightarrow 2x = 10 \Rightarrow x = 5$. Therefore, $y = 5$.
>
> Step 5: Maximum value $f(5,5) = 25$.
>
> **Answer: Max value = 25 at (5,5)**

LINEAR REGRESSION

[10 Marks]

Q: State the normal equations to fit a straight line $y = a + bx$ to a set of n data points (x_i, y_i).

> Step 1: The least squares method minimizes $\Sigma(y_i - a - bx_i)^2$.
>
> Step 2: Taking partial derivatives with respect to a and b and setting them to zero yields the normal equations.
>
> Step 3: Eq 1: $\Sigma y = n a + b \Sigma x$.
>
> Step 4: Eq 2: $\Sigma xy = a \Sigma x + b \Sigma x^2$.
>
> **Answer: $\Sigma y = na + b\Sigma x$; $\Sigma xy = a\Sigma x + b\Sigma x^2$**

LEAST APPROXIMATION METHOD

[10 Marks]

Q: Write the matrix equation for finding the least squares approximate solution to an overdetermined system $Ax = b$.

> Step 1: In an overdetermined system, $Ax = b$ may have no exact solution.
>
> Step 2: To minimize the error $||Ax - b||^2$, we multiply both sides by A^T (A transpose).
>
> Step 3: The normal equation becomes $(A^T A)x = A^T b$.
>
> Step 4: Solve for x: $x = (A^T A)^{-1} A^T b$.
>
> **Answer: $x = (A^T A)^{-1} A^T b$**
